



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
Mid Semester Examination 2024-25

Date of Examination: 18 /09/24 Session(FN/AN) FN Duration 2 hrs, Marks = 50
 Sub No: CS60075 Sub Name: Natural Language Processing
 Department/Centre/School: Computer Science and Engineering
 Specific charts, graph paper, log book etc. required NO
 Special Instructions (if any) ANSWER ALL questions. In case of reasonable doubt, make assumptions and state them upfront. Marks will be deducted for sketchy answers and claims without proper reasoning. All parts of a single question should be done at the same place. Keep probability values in fractional forms.

- Bigrams:** You are given the following data: vocabulary $V = \{w_1, w_2, w_3\}$ and the bigram probability distribution p on $V \times V$ given by: $p(w_1, w_1) = p(w_3, w_3) = \frac{1}{4}$, $p(w_2, w_2) = 0$, $p(w_2, w_1) = \frac{1}{8}$, $p(w_1, w_3) = \frac{1}{4}$, $p(w_1, *) = \frac{1}{2}$ (that is w_1 as the first of a pair), $p(*, w_2) = \frac{1}{8}$. Calculate $p(w_1, w_2)$, $p(w_3, w_2)$, $p(w_2, w_3)$ and $p(w_2|w_3)$. Show the steps of your calculation (no marks will be awarded unless the steps are clearly shown). [2.5 + 2.5 + 2.5 + 2.5 = 10]
- HMMs:** Given the sentence "I promise to back the bill." show how you would compute the probability of "back" as a verb versus the probability of "back" as a noun using the probabilities in Table 1 using the Viterbi algorithm. You are given the values for the third column of the Viterbi table which correspond to observation 3 or "to". They are VB: 0, TO: .00000018, NN: 0, PPSS: 0. Thus, you will show two computations both of which will use these values. Show the computation in each case. [3 + 3 = 6]

	<i>I</i>	<i>promise</i>	<i>to</i>	<i>back</i>
VB	0	.0093	0	.00008
TO	0	0	.99	0
NN	0	.0085	0	.00068
PPSS	.37	0	0	0

	VB	TO	NN	PPSS
<<	.019	.0043	.041	.067
VB	.0038	.035	.047	.0070
TO	.83	0	.00047	0
NN	.0040	.016	.087	.0045
PPSS	.23	.00079	.0012	.00014

Table 1: The observation and transition probabilities.

- Dependency trees:** Consider the following problems related to dependency trees.
 - You are given the following dependency relation tags: SUBJ-Subject, DOBJ-Direct object, PREP-Preposition, POBJ-Prepositional object, DET-Determiner. Draw all possible dependency parses (dependency trees with edges labelled with the dependency relation tags) for the sentence below.
The boy saw the girl with the telescope.
 - For this part use the tags in part (a) in addition if needed use ADJ-Adjective, PRED-Predicate. Show, in each case, by drawing a dependency tree whether the sentences (i) Heavy rain was forecast by the weatherman and (ii) Heavy rain was forecast for Friday by the weatherman result in projective or non-projective dependency trees. [4 + 2 + 2 = 8]
- Online learning:** Assume that you are learning a classifier for the data-driven deterministic parsing and the sentence I prefer ChatGPT course is a gold-standard parse in your training data. You are also given that ChatGPT and course are Nouns, I is a Pronoun while prefer is a Verb
 - Obtain the gold standard dependency graph for this sentence on your own.

- (b) Assume that your features correspond to the following 0/1 conditions:

C1: The stack is empty.

C2: Top of stack is **Noun** and head of buffer is **Verb**.

C3: Top of stack is **Pronoun** and head of buffer is **Verb**.

C4: The word at the top of stack occurs before word at the head of the buffer in the sentence.
Thus there are 4 conditions and 4 transitions (**LA**, **RA**, **SH**, **RE**) making the size of a feature vector = 16. Thus your feature vector will look like

[C1&LA,C2&LA,C3&LA,C4&LA | C1&RA,C2&RA,C3&RA,C4&RA |
C1&SH,C2&SH,C3&SH,C4&SH | C1&RE,C2&RE,C3&RE,C4&RE]

E.g., at some point, if stack top is empty and transition taken is **LA** then the feature vector will be:

[1,0,0,0 | 0,0,0,0 | 0,0,0,0 | 0,0,0,0]

The initial weight vector for **LA**, **RA**, **SH**, **RE** in that respective order is:

[2,2,2,2 | 3,3,3,2 | 2,2,2,2 | 2,2,2,2]

Use the gold standard parse obtained in part (a) during online learning. What will be the weights after completing two iterations of Arc-Eager parsing over this sentence. Show your steps clearly noting t^* , t_0 and the new weight vector each time as you proceed through the configurations.

[2 + 10 = 12]

5. Skip-grams: Recall that the skip-gram model is defined as follows

$$P(y|x) = \frac{\exp(\mathbf{v}_x \cdot \mathbf{c}_y)}{\sum_{y' \in \mathcal{V}} \exp(\mathbf{v}_x \cdot \mathbf{c}_{y'})}$$

where y is the context word, x is the center word, $\mathbf{v}$ and $\mathbf{c}$ are both d -dimensional vectors corresponding to the center and the context words. The skip-gram model considers the neighbors of a word to be words within a k -word window on either side (i.e., $k = 1$ gives the two immediately adjacent words). The skip-gram objective, log likelihood of this training data, is $\sum_{(x,y)} \log P(y|x)$, where the sum is over all training examples. Let the vocabulary size be $|\mathcal{V}|$.

- Given a *single* center word w and all possible context words $c \in \mathcal{V}$ in a given context position (say *immediate left*) what is the big-O running time to calculate $P(c|w)$? Report the answer in terms of d and $\mathcal{V}$.
- In skip-gram, we can efficiently approximate the denominator if for a given word we know what the largest terms are in its denominator. Suppose you are given a large corpus of text taken from the web. For a particular word w , how could you go about identifying 10 unique context words that are likely to have high weights in the denominator?
- What happens to the number of training examples if $k = 5$ instead of $k = 1$. Does this affect the runtime? How?
- Think about the context of the word balloons in the following sentences:
he blew up balloons for the birthday party
I popped balloons using a pin because I didn't like seeing the bright colors
In these contexts (and more generally), what do you think will change about what the skip-gram model learns about the *linguistic surrounding* of balloons as you change $k = 1$ to $k = 3$? What about $k = 10$?

[2 + 3 + 3 + 3 = 11]

6. Danish numbers: Of all the national languages of Europe, Danish has a bizarre vocabulary of number words. Here are some numbers and the corresponding number words in Danish: fire=4, nioghalvfjerds=79, toogtyve=22, seksogtres=66, ni=9, syvoghalvtreds=57, enogfirs=81, tre=3, fem=5. Fill in the blanks: seks: _____, nioghalvtreds: _____, treogtyve: _____. [3]